

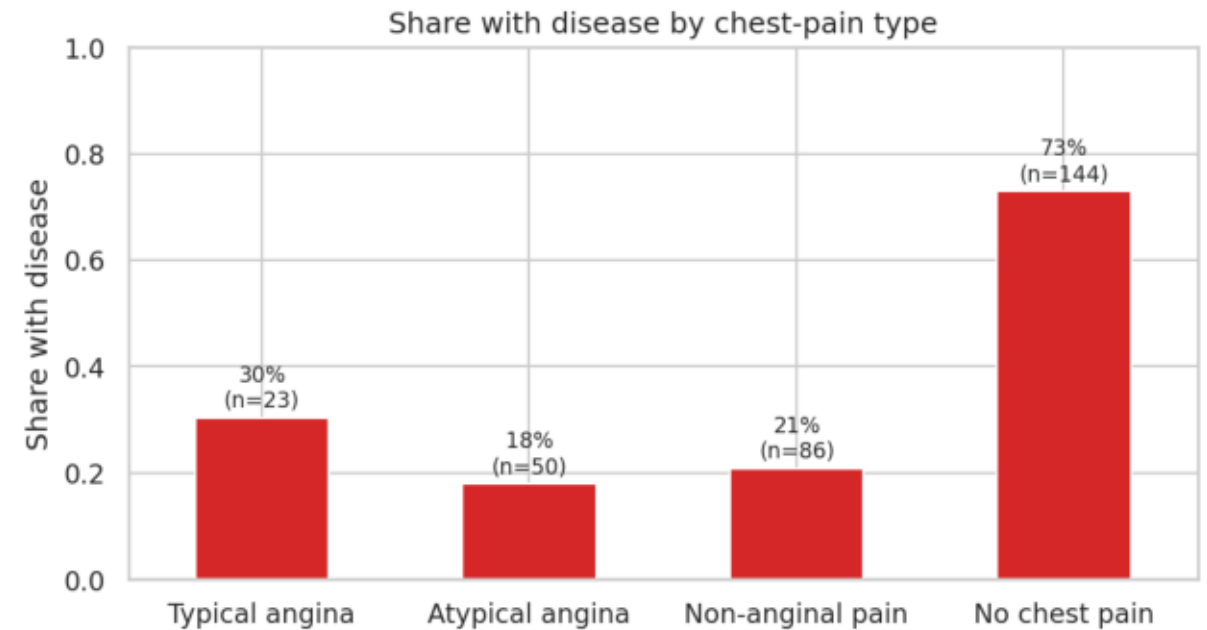
1. The question

- Client: a community clinic.
- Need: flag patients who should see a cardiologist; the tool does NOT diagnose.
- Target: heart disease (1) vs none (0); positive class = disease.
- Metric: recall ≥ 0.85 , because a missed sick patient can die, while a false alarm costs one extra visit.

- UCI Heart Disease (Cleveland): 303 patients × 14 columns.
- Biggest problem: '?' in ca and thal made pandas read them as text.
- Fix: read '?' as missing, convert to numbers, fill with the most frequent value (training data only).
- cp, thal, slope, restecg are categories stored as numbers → one-hot encoded.

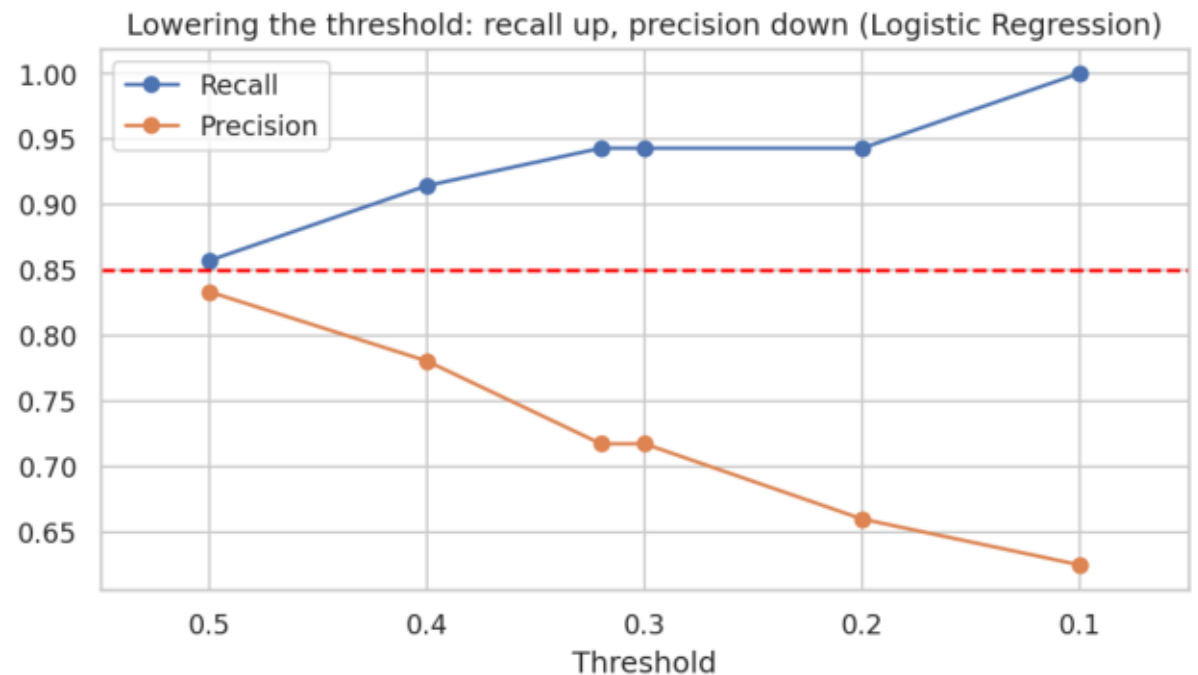
3. Key insights

- Patients with NO chest pain have the highest disease rate: 73%.
- Chest pain during exercise: 77% vs 31%.
- Risk climbs with narrowed vessels: 26% (0) → 85% (3).



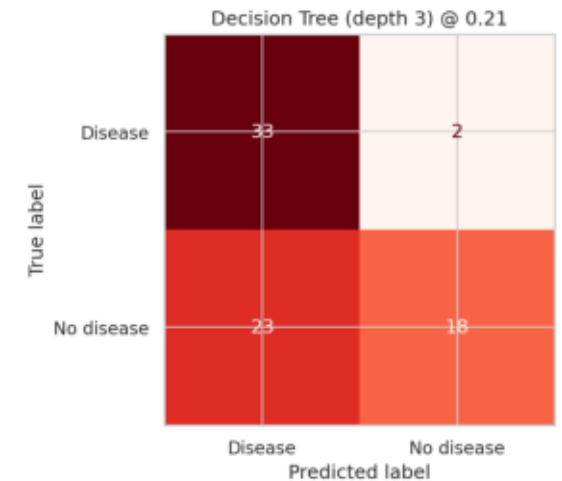
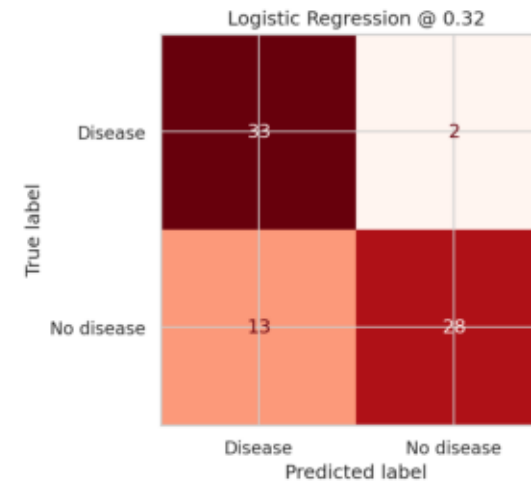
4. The model

- Baseline rule (no chest pain OR exercise angina): recall 0.80, fails the target.
- Logistic Regression: ROC AUC 0.94; decision tree: 0.83.
- Threshold chosen on training data (cross-validation) to reach recall $\geq 0.85 \rightarrow 0.32$.
- Why Logistic Regression: best ranking of patients and explainable odds.



5. Results

- Test recall 0.94: 33 of 35 sick patients referred, 2 missed.
- Price: precision 0.72 → 13 healthy patients referred (was 6).
- The 2 missed patients had high ST depression → add a safety rule: refer if oldpeak ≥ 2 .



6. Recommendation

- Use as a referral aid only, at threshold ≈ 0.32 , plus the oldpeak safety rule.
- Validate on the clinic's own patients and separately for women first.
- Limits: 303 patients, 1980s US data, CV recall 0.85 ± 0.10 , needs imaging for ca/thal.
- Never use it to diagnose, rule out disease or deny care.